Peiwen Zhou 周培文

Curriculum Vitae

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August 2005 | Yangzhou, Jiangsu

Education

China Agricultural University — Beijing, China
B.S. in Mathematics and Applied Mathematics — Aug. 2023 – Jul. 2024; Sep. 2025 – present
Expected graduation: June 2028 GPA: 3.73/4.00 Rank 21/80

University of Wisconsin–Madison — Madison, WI, USA
Visiting Student GPA: 3.90/4.00 — Aug. 2024 – Jul. 2025

Research Experience

Exit-Angle Selection and Collision-Manifold Routing in the Anisotropic Kepler Problem
Advisor: Prof. Dallas Albritton — Feb. 2026 – present
Collaborators: Andrew R. Kulik and Gaorui Yan
Research on the anisotropic Kepler problem and its numerical simulation.

Pattern Formation in a Dual-Fractional Vegetation–Sand Model
Advisor: Prof. Feifan Zhang — Oct. 2025 – present
Study of pattern formation in arid ecosystems using a dual-fractional vegetation–sand model.

Regularization and Collapse in the Two-Body Problem
Advisor: Prof. Dallas Albritton — Summer 2025
Supported by NSF grants DMS-2406947 and DMS-2037851.

Publications and Manuscripts

1. **A Study on the Effects of Riesz Fractional Diffusion on Pattern Formation in a Toxic-Phytoplankton–Zooplankton Model**
   Ying Gong, Yuhan Yan, **Peiwen Zhou**, Lei Zhao, and Feifan Zhang.
   Mathematics, February 2026.
2. **Pattern formation and propagation in a nonlocal wind-driven vegetation–sand model**
   Peiwen Zhou, Zimo Liu, Huihui Pang, and Feifan Zhang.
   Submitted.

Honors and Awards

Second Prize, 11th China University Life Sciences Competition — Aug. 2026
First Prize, 36th Beijing College Mathematics Competition — Nov. 2025
Third Prize, 17th China College Mathematics Competition — Nov. 2025
Dean's List, University of Wisconsin–Madison — Dec. 2024; May 2025

Skills and Selected Coursework

Programming and tools: Python, MATLAB, and LaTeX
Languages: Chinese; English (fluent); Japanese (beginner); Cantonese (beginner)
Duolingo English Test: 130 **College English Test Band 6 (CET-6): 558**
Selected coursework: Ordinary Differential Equations; Partial Differential Equations; Modern Algebra; Probability Theory; Complex Analysis; Optimization